

Cassie Charlie

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EDUCATION

National Dong Hwa University (國立東華大學)

Hualien, Taiwan

Bachelor of Computer Science and Information Engineering

Graduation Date: Jun 2026

WORK EXPERIENCE

National Dong Hwa University

Hualien, Taiwan

Research Assistant at Prof. Wen-Chieh Fang's Lab

Sep 2025 - Mar 2026

- Conducted ablation studies and hyperparameter optimization for diffusion-based speech enhancement, achieving 38.6% CSIG improvement and 16.6% STOI gain over baseline configurations.
- Analyzed model performance across 6 evaluation metrics (PESQ, CSIG, CBAK, COVL, SSNR, STOI) to identify optimal architectures for speech enhancement systems.
- Developed chunked inference pipeline with overlap blending, enabling 94% ASV pass rate while maintaining audio quality for full-length samples.

National Dong Hwa University

Hualien, Taiwan

Teaching Assistant | Intro to Machine Learning

Oct 2025 - Jan 2026

- Managed a classroom of 45 students in equipment setup, attendance tracking, and classroom support for hands-on ML sessions.
- Developed homework solutions and grading rubrics; evaluated 150+ assignments and exams with consistent feedback and fair assessment.
- Conducted office hours and review sessions, helping students debug Python code and understand core ML concepts.

PROJECT EXPERIENCE

National Science and Technology Council

Hualien, Taiwan

EmoScale | NSTC-Funded Undergraduate Research

Jul 2025 - Mar 2026

- Processed and analyzed speech datasets using OpenSMILE feature extraction to model emotional intensity relationships across audio samples.
- Architected embedding extraction pipeline (HuBERT content, ECAPA-TDNN speaker, Wav2Vec2 emotion) supporting diffusion decoder training for 6 speakers with zero-shot evaluation on 2 unseen speakers.
- Evaluated model performance using RMSE analysis across intensity scales (0.0–1.0) to measure system responsiveness to conditioning signals.

Personal Project

Hotel Revenue Analysis

Mar 2025 - Apr 2025

- Analyzed 3 years of hotel data using SQL (JOINS, GROUP BY, UNION) to compute revenue, ADR, and distribution by hotel type.
- Built interactive Power BI dashboard using DAX measures and KPI indicators to visualize revenue trends and hotel performance metrics.

SKILLS & LANGUAGES

Skills: • Microsoft Office (Excel, Word, PowerPoint)

- Data Analytics: SQL, Power BI, Excel
- Programming Languages: Python (Pandas, NumPy, Matplotlib)

Languages: Indonesian (Native), English (Fluent, IELTS 8.0), Chinese (Conversational), Korean (Conversational)